

# AI Resume Bullet Improver

## Introduction

AI Resume Bullet Improver is an AI-powered web application that helps users transform simple resume bullet points into professional, ATS-friendly achievements. The application leverages Artificial Intelligence to rewrite resume content using strong action verbs, improved grammar, and industry-standard formatting.

Built using HTML, CSS, JavaScript, and Vercel Serverless Functions, the application provides a secure and modern solution for integrating AI without exposing API credentials. The intuitive interface, responsive design, and real-time AI assistance make it a valuable tool for students, freshers, and professionals preparing job-ready resumes.

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## Project Overview

The primary goal of this project is to simplify resume writing by using Artificial Intelligence to enhance resume bullet points. Users can enter a basic description of their work, select their preferred writing style, experience level, and technology stack, and instantly receive a professionally rewritten resume bullet.

The project demonstrates modern frontend development practices combined with secure AI integration through serverless architecture.

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## Objectives

- Develop a modern AI-powered resume improvement application.
  - Generate professional and ATS-friendly resume bullet points.
  - Provide multiple writing styles for different professional needs.
  - Improve resume readability using strong action verbs.
  - Implement secure AI integration using Vercel Serverless Functions.
  - Build a responsive and user-friendly interface.
  - Demonstrate secure API architecture without exposing API keys.
  - Showcase practical AI integration using Groq AI.
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# Features

## AI Resume Improvement

- Professional resume bullet rewriting
- ATS-friendly optimization
- Strong action verb enhancement
- Improved grammar and readability

## Resume Customization

- Writing Style Selection
- Experience Level Selection
- Technology Stack Selection

## User Experience

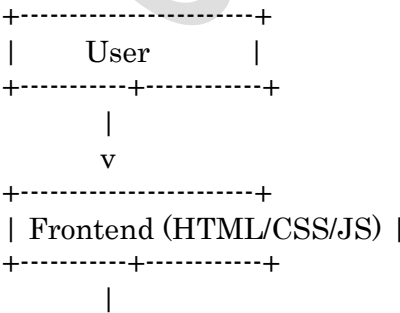
- Responsive Design
- Modern Glassmorphism UI
- Dark & Light Theme
- Mobile Navigation
- Toast Notifications
- Copy to Clipboard
- Local Storage Support
- Real-time Word Counter
- Real-time Character Counter

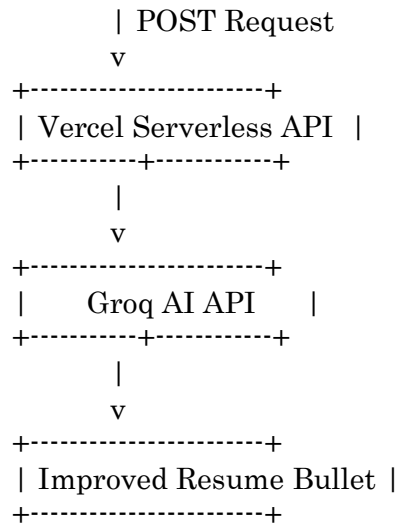
## AI Integration

- Secure Serverless API
- Groq AI Integration
- Prompt Engineering
- Fast AI Response

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# System Architecture





The frontend communicates with a secure serverless backend, which processes AI requests using the Groq API. This architecture ensures that API credentials remain protected while delivering fast AI-generated responses.

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## Technologies Used

### Frontend

- HTML5
- CSS3
- JavaScript (ES6+)
- CSS Variables
- Fetch API
- Local Storage API
- Font Awesome
- Google Fonts (Inter)

### Backend

- Vercel Serverless Functions
- Node.js

### Artificial Intelligence

- Groq AI API
- Llama 3.3 70B Versatile

## Development Tools

- Visual Studio Code
  - Git
  - GitHub
  - Vercel
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## Database Design

This application does not use a traditional database.

Instead, it utilizes:

### Local Storage

Used for storing:

- Theme Preference
- Resume Form Data
- Previous AI Result

### Environment Variables

Used for securely storing:

- Groq API Key

The serverless function securely accesses the API key without exposing it to the frontend.

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## Project Workflow

User Opens Application



Enter Resume Bullet

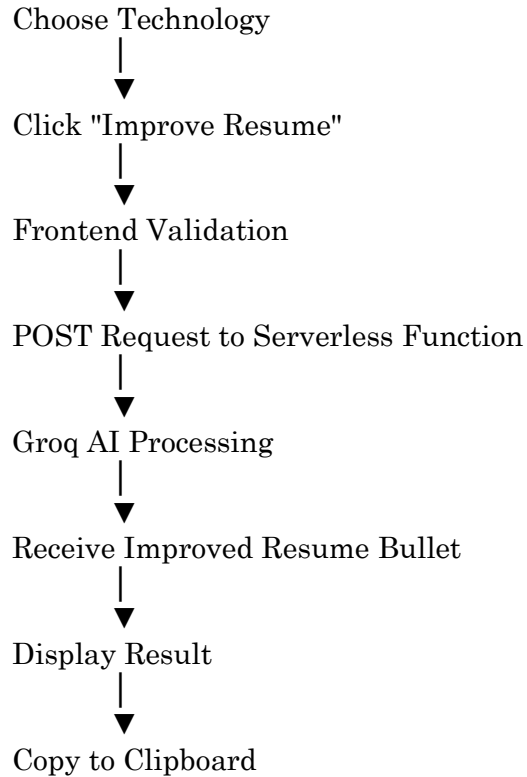


Choose Writing Style



Choose Experience Level





## Implementation Details

### Frontend Development

Designed a clean and responsive user interface using semantic HTML5, modern CSS3, and vanilla JavaScript.

### Responsive Design

Implemented responsive layouts for desktop, tablet, and mobile devices using CSS Grid, Flexbox, and media queries.

### User Interface

Developed modern UI components including:

- Responsive Navigation
- Hero Section
- Resume Improvement Form
- Result Card
- Feature Cards
- Professional Footer

## AI Communication

Implemented asynchronous communication using the Fetch API to send resume data securely to a Vercel Serverless Function.

## Serverless Backend

Created a serverless API endpoint that:

- Receives frontend requests
- Validates user input
- Generates AI prompts
- Communicates with Groq AI
- Returns optimized resume bullets

## Security

Protected sensitive API credentials using:

- Environment Variables
- Serverless Functions

No API keys are exposed to the client.

## User Experience Enhancements

- Theme Toggle
  - Toast Notifications
  - Loading State
  - Clipboard Support
  - Local Storage Persistence
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## Learning Outcomes

Through this project, the following concepts were learned and implemented:

- Responsive Web Design
- Modern JavaScript (ES6+)
- DOM Manipulation
- Fetch API
- Async/Await
- REST API Communication
- Prompt Engineering
- AI Integration

- Serverless Functions
  - Environment Variables
  - Local Storage
  - Clipboard API
  - Error Handling
  - Secure API Architecture
  - Modern UI/UX Design
  - Deployment using Vercel
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## Conclusion

AI Resume Bullet Improver successfully demonstrates how Artificial Intelligence can simplify resume writing by transforming basic resume statements into professional, ATS-friendly achievements.

The project combines modern frontend technologies with secure serverless backend architecture to provide a fast, responsive, and secure user experience. It highlights practical AI integration, responsive web development, and deployment best practices while ensuring sensitive API credentials remain protected.

This project serves as an excellent portfolio application that showcases real-world AI implementation, modern JavaScript development, secure API communication, and cloud deployment.

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## Project Links

GitHub Repository

<https://github.com/charanepuri/ai-resume-bullet-improver>

Live Demo

<https://ai-resume-bullet-improver.vercel.app/>

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## Author

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### **Django Portfolio**

<https://portfolio-site-django.onrender.com>

### **GitHub**

<https://github.com/charanepuri>

### **React Portfolio**

<https://charan-react-portfolio.vercel.app>

### **LinkedIn**

<https://www.linkedin.com/in/charan-teja-972aa9231>

### **Flask Portfolio**

<https://flask-developer-dashboard-portfolio.onrender.com/>